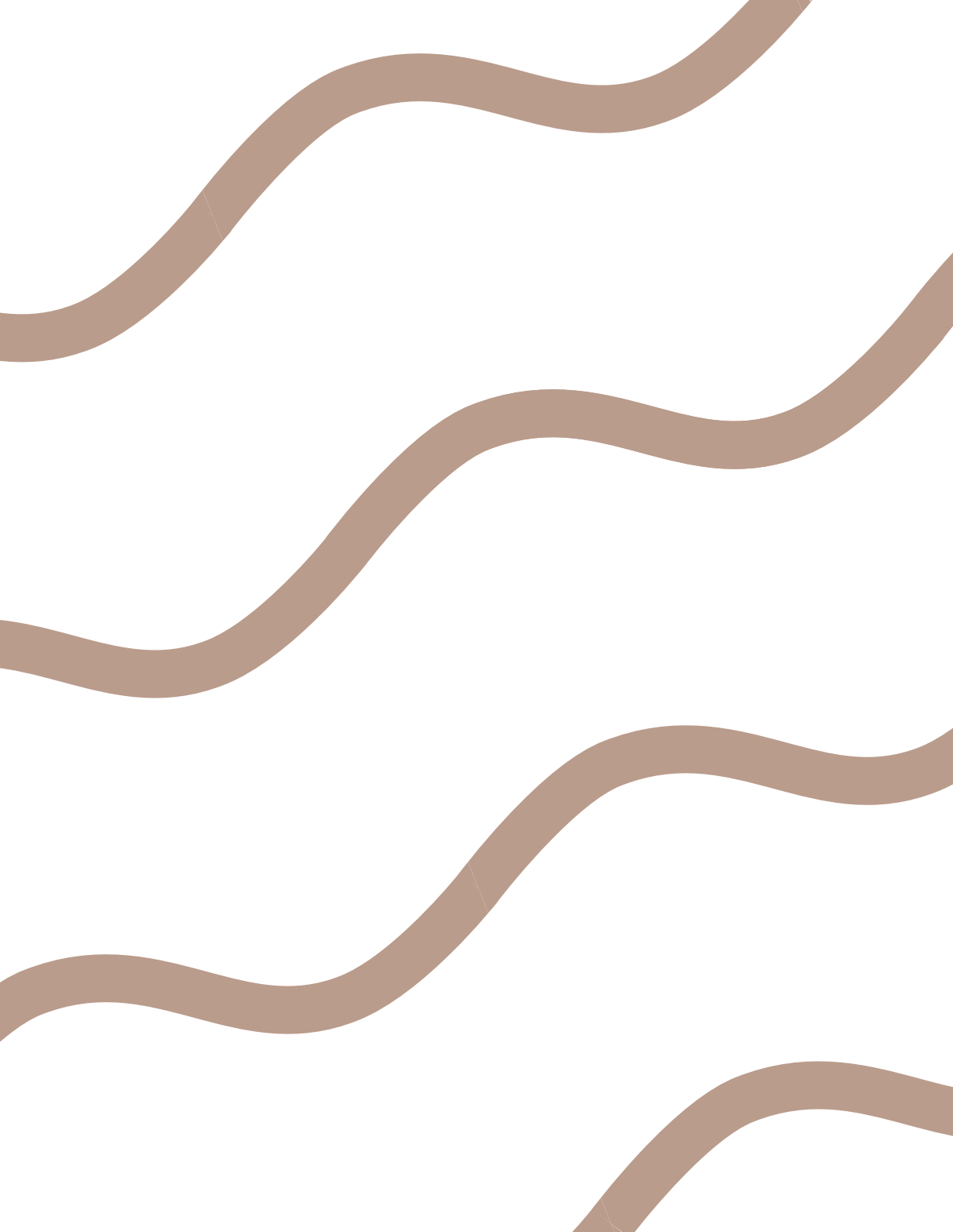




(5) 设 D 是以 $(1,1)$, $(-1,1)$ 和 $(-1,-1)$ 为顶点的三角形区域, D_1 是 D 在第一象限的部分, 且 $f(x,y)$

$$= xy + \iint_D f(x,y) dx dy, \text{ 其中 } f(x,y) \text{ 在 } D \text{ 上连续, 则 } ().$$

A. $\iint_D f(x,y) dx dy = \iint_{D_1} f(x,y) dx dy$

B. $\iint_D f(x,y) dx dy = 2 \iint_{D_1} f(x,y) dx dy$

C. $\iint_D f(x,y) dx dy = \iint_D f(y,x) dx dy$

D. $\iint_D f(x,y) dx dy = 2 \iint_{D_1} f(y,x) dx dy$

(2) 设 D 是由曲线 $|x|y + |x| + y = 1$ 与 x 轴所围成的有界区域, 计算 $I = \iint_D [2\ln(1+y) - y + x] dx dy$.

$$\int_0^1 x^2 \sqrt{2x-x^2} dx = \int_0^1 x^2 \sqrt{1-(x-1)^2} dx$$

$$\stackrel{x-1=\sin t}{=} \int_{-\frac{\pi}{2}}^0 (1+\sin)^2 \cdot \cos dt$$

$$\stackrel{t=-u}{=} \int_{\frac{\pi}{2}}^0 (1-\sin u) \cdot \cos u du$$

(13) 设 $f(x,y)$ 在区域 $0 \leq x \leq 1, 0 \leq y \leq 1$ 上连续, $f(0,0) = 0$, 且 $f(x,y)$ 在点 $(0,0)$ 处可微, $f'_y(0,0) =$

$$1. \text{ 求 } \lim_{x \rightarrow 0^+} \frac{\int_0^x dt \int_t^1 f(t,u) du}{1 - e^{-\frac{x^2}{4}}}.$$

(4) 积分 $I = \int_0^{2\pi} d\theta \int_{\frac{\pi}{2}}^{\pi} \theta^2 e^{r^2} dr =$

(10) 设 $D = \{(x,y) | 0 \leq y \leq 1-x, 0 \leq x \leq 1\}$, 则 $\iint_D e^{\frac{x}{1+y}} dx dy =$

6

21. (本题满分 12 分)

设平面区域 $D = \{(x, y) \mid x + y \leq 1, x \geq 0, y \geq 0\}$, 计算 $\iint_D \frac{e^{-(x+y)}}{\sqrt{xy}} d\sigma$.

7

(7) 设曲线 $L: y = \cos x$, L_1 与 L_2 分别为 L 上点 $(-\frac{\pi}{2}, 0)$ 与 $(\frac{\pi}{2}, 0)$ 处的切线, 若函数 $f(x, y)$ 在

L, L_1 与 L_2 所围闭区域 D 上连续, 则下列等式成立的有 () 个.

- ① $\iint_D f(x, y) d\sigma = \int_{-\pi/2}^{\pi/2} dx \int_{\cos x}^{\pi/2} f(x, y) dy$
 ② $\iint_D f(x, y) d\sigma = \int_0^{\pi/2} dy \int_{y-\pi/2}^{\pi/2-y} f(x, y) dx$
 ③ $\iint_D f(x, y) d\sigma = 2 \int_0^{\pi/2} dy \int_{\arccos y}^{\pi/2-y} f(x, y) dx + \int_{-1}^1 dy \int_{y-\pi/2}^{\pi/2-y} f(x, y) dx$
 ④ $\iint_D f(x, y) d\sigma = \int_{-\pi/2}^0 dx \int_{\cos x}^{\pi/2+x} f(x, y) dy + \int_0^{\pi/2} dx \int_{\cos x}^{\pi/2-x} f(x, y) dy$
 (A) 1 (B) 2 (C) 3 (D) 4

8

20. (本题满分 12 分)

设 $f(x, y)$ 连续, 且

$$f(x, y) = \frac{1}{\sqrt{x^2 + y^2 - 2}} + \frac{1}{\pi} \iint_D f(x, y) d\sigma,$$

其中 $D = \{(x, y) \mid 0 \leq y \leq \sqrt{2x - x^2}\}$, 试求 $\iint_D f(x, y) d\sigma$.

(5)C.

解 由已知,三角形区域 D 如图 5-34 所示.

记 $A = \iint_D f(x, y) dx dy$, 则 $f(x, y) = xy + A$. 等式两边同时在 D 上积分, 得

$$\iint_D f(x, y) dx dy = \iint_D xy dx dy + \iint_D A dx dy,$$

故 $A = \iint_D xy dx dy + 2A$.

作辅助线 $y = -x$, 将 D 分成 D_1 与 D_2, D_3 与 D_4 , 则

$$\iint_D xy dx dy = \iint_{D_1 \cup D_2} xy dx dy + \iint_{D_3 \cup D_4} xy dx dy,$$

$D_1 \cup D_2$ 关于 y 轴对称, xy 关于 x 是奇函数, $D_3 \cup D_4$ 关于 x 轴

对称, xy 关于 y 是奇函数, 故 $\iint_{D_3 \cup D_4} xy dx dy = 0$, 从而 $A = 0 + 2A$,

即 $A = 0$. 故 $f(x, y) = xy = yx$, 选项 C 正确.

选项 A, B, D 中等号左端均为零, 但右端不为零, 排除选项 A, B, D.

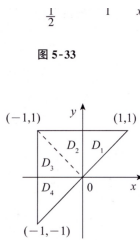
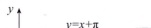


图 5-33

图 5-34



(2) 以 $-x$ 代替 x , 方程 $|x| + |x| + |y| = 1$ 不变, 可知曲线关于 y 轴对称. 当 $x \geq 0$ 时, 由 $|x| + |x| + |y| = 1$, 得 $y = \frac{1-x}{1+x}$, 即 $y = 1 - \frac{2x}{1+x}$. 它是关于 $x = -1, y = -1$ 为渐近线的双曲线. 有界区域 D 如图 5-42 阴影所示.

$$I = \iint_D [2\ln(1+y) - y + x] dx dy = \int_0^1 \int_{-1}^0 [2\ln(1+y) - y] dx dy + \int_0^1 x dy$$

对于 I_1, x 是奇函数, 于是 $I_1 = 0$.

对于 $I_2, 2\ln(1+y) - y = [2\ln(1+y) - y] x^2$, 它是关于 x 的偶函数.

故, 数.

$$I_2 = 2 \int_0^1 [2\ln(1+y) - y] dy \quad (D_1 \text{ 是 } D \text{ 在第一象限的部分}).$$

考虑到被积函数的特性, 先对 x 积分, 后对 y 积分. 由 $xy + x + y = 1$, 知 $x = \frac{1-y}{1+y}$, 则

$$I_2 = 2 \int_0^1 [2\ln(1+y) - y] dy = 2 \int_0^1 [2\ln(1+y) - y] \cdot \frac{1-y}{1+y} dy.$$

注意: $[2\ln(1+y) - y] = \frac{2}{1+y} - 1 = \frac{1-y}{1+y}$, 故

$$I_2 = 2 \int_0^1 [2\ln(1+y) - y] \cdot \frac{1-y}{1+y} dy = 2 \times \frac{1}{2} [2\ln(1+y) - y]^2 \Big|_0^1 = (2\ln 2 - 1)^2.$$

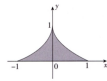


图 5-42

(13) 解 交换积分顺序, 如图 5-50 所示.

$$\int_0^1 dx \int_x^1 f(t, u) du = - \int_0^1 du \int_0^u f(t, u) dt$$

$$\text{故原式} = \lim_{x \rightarrow 1^+} - \int_0^1 du \int_0^u f(t, u) dt \quad \text{洛必达} \quad \lim_{x \rightarrow 1^+} - \int_0^1 f(t, x) dt$$

由积分中值定理, 得

$$\int_0^1 f(t, x) dt = f(\xi, x) x^2 \quad (0 \leq \xi \leq x^2),$$

$$\text{故原式} = \lim_{x \rightarrow 1^+} - \frac{f(\xi, x) x^2}{x^3}.$$

因 $f(x, y)$ 在点 $(0, 0)$ 处可微, 由可微的定义, 有

$$f(\xi, x) = f(0, 0) + f'_x(0, 0)\xi + f'_y(0, 0)x + o(\sqrt{x^2 + \xi^2}).$$

$$\text{又} \left| \frac{f'_x(0, 0)\xi}{x} \right| \leq \left| \frac{f'_x(0, 0)x^2}{x} \right| = |f'_x(0, 0)x|, \text{ 则有}$$

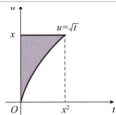


图 5-50

$$(4) \frac{4}{3} e^{\pi^2} (\pi^2 - 1) + \frac{4}{3}.$$

解 积分区域 D 如图 5-37 所示, 交换极坐标顺序, 则

$$I = \int_0^{\pi} d\theta \int_0^{\frac{\pi}{2}} \theta^2 e^{\theta^2} d\theta = \int_0^{\pi} e^{\theta^2} \cdot \frac{1}{3} \theta^3 \Big|_0^{\frac{\pi}{2}} d\theta = \int_0^{\pi} e^{\theta^2} \cdot \frac{8}{3} \theta^2 d\theta$$

$$= \frac{8}{3} \int_0^{\pi} \theta^2 e^{\theta^2} d\theta = \frac{4}{3} \int_0^{\pi} t e^t dt = \frac{4}{3} e^t (t - 1) \Big|_0^{\pi} = \frac{4}{3} e^{\pi^2} (\pi^2 - 1) + \frac{4}{3}.$$

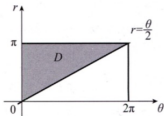


图 5-37

李林金家梅之②

2026 考研数学精读精练 880 题 (数学二) 解析分析

$$\lim_{x \rightarrow 0^+} \frac{f'_x(0, 0)\xi}{x} = 0, \quad \lim_{x \rightarrow 0^+} \frac{o(\sqrt{x^2 + \xi^2})}{x} = \lim_{x \rightarrow 0^+} \frac{o(x)}{x} = 0,$$

$$\text{故} \lim_{x \rightarrow 0^+} - \frac{f(\xi, x) x^2}{x^3} = \lim_{x \rightarrow 0^+} \frac{f(0, 0) + f'_x(0, 0)\xi + f'_y(0, 0)x + o(\sqrt{x^2 + \xi^2})}{-x} = -f'_y(0, 0) = -1.$$

(10) $\frac{1}{2}(e-1)$.

解 积分区域 D 如图 5-13 所示, 采用极坐标.

$$\begin{aligned} \iint_D e^{\frac{1}{\cos \theta}} dx dy &= \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\cos \theta}{\cos \theta + \sin \theta}} e^{\frac{1}{\cos \theta}} r dr \\ &= \int_0^{\frac{\pi}{2}} e^{\frac{1}{\cos \theta}} \cdot \frac{1}{2} r^2 \Big|_0^{\frac{\cos \theta}{\cos \theta + \sin \theta}} d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} e^{\frac{1}{\cos \theta}} \cdot \frac{1}{(\cos \theta + \sin \theta)^2} d\theta \\ &= -\frac{1}{2} \int_0^{\frac{\pi}{2}} e^{\frac{1}{\cos \theta}} d(\cos \theta + \sin \theta) \\ &= -\frac{1}{2} e^{\frac{1}{\cos \theta}} \Big|_0^{\frac{\pi}{2}} = \frac{1}{2}(e-1). \end{aligned}$$

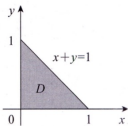
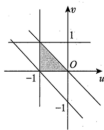


图 5-13

法二 用换元法.

令 $\begin{cases} u = -(x+y), \\ v = x, \end{cases}$ 则 $D' = \{(u, v) \mid -1 \leq u \leq 0, 0 \leq v \leq 1, -1 \leq u+v \leq 0\}$, 区域 D' 如图中阴影部分所示.



$$\begin{aligned} \text{原式} &= \iint_D \frac{e^x}{\sqrt{v(-v-u)}} \left| \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} \right| du dv = \iint_{D'} \frac{e^u}{\sqrt{v(-v-u)}} du dv \\ &= \int_{-1}^0 du \int_0^{-u} \frac{e^u}{\sqrt{-v^2 - uv}} dv = \int_{-1}^0 e^u \left[\arcsin \frac{v + \frac{u}{2}}{-\frac{u}{2}} \right]_0^{-u} du \\ &= \int_{-1}^0 e^u \left(\frac{\pi}{2} + \frac{\pi}{2} \right) du = \int_{-1}^0 \pi e^u du = \pi \left(1 - \frac{1}{e} \right). \end{aligned}$$

(7) 答案 选(A).

解 由题意知 $L_1: y = x + \frac{\pi}{2}$, $L_2: y = -x + \frac{\pi}{2}$. ①, ②, ③不成立, ④成立. 故选(A).

20. 解 令 $A = \iint_D f(x, y) d\sigma$, 则由题设得

$$f(x, y) = \frac{1}{\sqrt{x^2 + y^2 - 2}} + \frac{1}{\pi} A, \quad \cdots \cdots 3 \text{分}$$

两边同时积分, 得

$$\iint_D f(x, y) d\sigma = \iint_D \frac{1}{\sqrt{x^2 + y^2 - 2}} d\sigma + \iint_D \frac{1}{\pi} A d\sigma,$$

即

$$A = \iint_D \frac{1}{\sqrt{x^2 + y^2 - 2}} d\sigma + \frac{A}{\pi} \cdot \frac{1}{2} \pi,$$

亦即

$$A = 2 \iint_D \frac{1}{\sqrt{x^2 + y^2 - 2}} d\sigma. \quad \cdots \cdots 5 \text{分}$$

如图所示, 曲线 $x^2 + y^2 = 2$ 将区域 D 分成 D_1, D_2 两部分, 则

$$\begin{aligned} A &= 2 \iint_{D_1} \frac{1}{\sqrt{x^2 + y^2 - 2}} d\sigma \\ &= 2 \iint_{D_1} \frac{1}{\sqrt{x^2 + y^2 - 2}} d\sigma + 2 \iint_{D_2} \frac{1}{\sqrt{x^2 + y^2 - 2}} d\sigma \quad \cdots \cdots 8 \text{分} \\ &= 2 \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sqrt{2}} \frac{1}{\sqrt{4 - r^2}} r dr + 2 \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta \int_0^{\sqrt{2}} \frac{1}{\sqrt{4 - r^2}} r dr + 2 \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} d\theta \int_0^{\sqrt{2}} \frac{1}{\sqrt{4 - r^2}} r dr \\ &= (2 - \sqrt{2})\pi. \quad \cdots \cdots 12 \text{分} \end{aligned}$$